

NAGA LAVANYA MUKKU

nagalavanyamukku@gmail.com | +1 (773) 332-8587 | linkedin.com/in/lavanyamukku

SUMMARY

Software Engineer with 4+ years of experience delivering scalable full-stack applications across frontend and cloud-native environments. Proficient in React.js, TypeScript, Node.js, PostgreSQL, MongoDB, REST APIs, AWS, Docker, and Kubernetes. Demonstrated expertise in responsive UI development, scalable backend services, microservices architecture, CI/CD pipelines, and Agile-based SDLC workflows.

EXPERIENCE

Kalyan Jewellers Inc., Chicago, IL -- Software Engineer

Dec 2024 – Present

- Developed scalable RESTful APIs using **Node.js**, **Express.js**, **PostgreSQL**, and **JWT authentication** for enterprise retail order management pipelines handling 5,000+ daily transactions.
- Built responsive **React.js** and **TypeScript** frontend applications using **Redux** state management, reusable UI components, and REST API integrations, reducing frontend load times by 30%.
- Implemented real-time event-driven communication systems using **WebSockets** and **Socket.io** for distributed order tracking platforms, reducing synchronization latency to under 2 seconds.
- Optimized **Node.js** backend services, **PostgreSQL** transaction operations, and **Nginx** load balancing configurations, improving application scalability and high-availability during peak traffic periods.
- Integrated **Docker** containerization, **GitHub Actions CI/CD pipelines**, and **AWS deployment** solutions for cloud-native deployment automation and production observability.
- Leveraged **GitHub Copilot** and **OpenAI-assisted** development tools for debugging, API integrations, and frontend feature development, accelerating software delivery cycles by 30%.

Tata Consultancy Services (TCS), Chennai, India -- Associate Software Developer

Jun 2019 – Jul 2022

- Developed enterprise financial reporting services using **Node.js**, **Express.js**, and **MongoDB** for large-scale analytics systems supporting 500+ business stakeholders.
- Designed interactive **React.js** and **TypeScript** reporting dashboards with reusable UI components, **Redux** state management, and dynamic data visualization features for enterprise business intelligence platforms.
- Optimized large-scale financial data processing workflows, **MongoDB aggregation pipelines**, and backend reporting services, improving reporting system performance by 35%.
- Implemented **Jest** and **Cypress** automated testing frameworks across **React.js** frontend modules and **Node.js** backend services, improving production release reliability and application stability.
- Refactored legacy **TypeScript** application modules and **REST API** integrations using scalable component-driven development standards, reducing production defects across multiple release cycles.
- Collaborated within **Agile Scrum** teams on production support, release management, **Git-based version control**, and cross-functional enterprise application delivery.

PROJECTS

Real-Time Order Tracking System

- Built a real-time retail order tracking platform using React.js, TypeScript, Node.js, WebSockets, PostgreSQL, and Redis with scalable REST APIs and event-driven communication applications.

Financial Reporting Dashboard

- Developed enterprise financial reporting dashboards using React.js, TypeScript, Node.js, MongoDB, and REST APIs with dynamic data visualization and optimized data processing workflows.

EDUCATION

Pace University, New York, United States- *M.S., Computer Science, Graduated 2024*
Jawaharlal Nehru Technological University, India – *B.Tech., Electronics and Communication Engineering, Graduated 2019*

TECHNICAL SKILLS

Programming Languages	: JavaScript (ES6+), TypeScript, HTML5, CSS3, SQL
Frontend Technologies	: React.js, Redux, React Hooks, Tailwind CSS, Bootstrap, Webpack, Vite
Backend Technologies	: Node.js, Express.js, REST APIs, GraphQL, JWT Authentication, RBAC, Microservices
Testing & Monitoring	: Jest, Cypress, React Testing Library, AWS CloudWatch, ELK Stack
Tools & Workflow	: Git, Jira, Agile/Scrum
Cloud & DevOps	: AWS (EC2, S3, Lambda, EKS), Docker, Kubernetes, GitHub Actions, CI/CD, Nginx, Linux
Databases	: PostgreSQL, MongoDB, Redis